Speed Competition and Strategic Trading*

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This paper examines the effect of speed competition on price informativeness in financial markets with large strategic traders. Speed competition renders fast traders to trade earlier and temporally fragments the price discovery process. Large traders internalize their price impact and strategically shade trading aggressiveness to resolve the pre-trading uncertainty proportional to the number of traders in each trading round. Therefore, the price discovery in the late period depends on not only the number of slow traders but also the amount of fundamental uncertainty resolved earlier from fast traders. A concentration of fast or slow traders harms price discovery in the late period, generating hump-shape overall price informativeness to speed competition. With fast information diffusion, speed competition harms the overall price informativeness. However, when fast traders are high-frequency traders, speed competition benefits the overall price informativeness.

JEL classification: *G12, G14, G18, D41, D44, D47*
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1. Introduction

An extensive literature in economics and finance argues that private information is transmitted into asset prices through the trading of informed traders. A canonical framework of this process is given by Kyle (1985). Knowing the fundamental value of an asset and internalizing the price impact of trades, informed traders cautiously shade trading aggressiveness based on their private information. The vast literature building on this framework provides important insights about market outcomes in various market settings, which guide a large body of empirical analysis and policy recommendations.[1] Considering the advancement of speed technology, one of the most salient phenomena in securities trading in recent years,[2] the current literature is largely abstract from examining how the speed hierarchy, which temporarily fragments the progress of information aggregation, influences price informativeness.

Intensified speed competition brings a certain amount of information revealed late to early, which naturally affects information aggregation and price discovery in the short and long runs. The overall effect depends on whether price discovery has increasing or decreasing returns to scale with competitive or strategic traders.[3] The existing literature, such as Kendall (2018), Dugast and Foucault (2018) and Huang and Yueshen (2021), focuses on the price informative consequence of intensified speed competition with small competitive traders. Empirical evidence shows the existence of anticipatory trading of fast traders who are, however, large and informed and trade strategically (Hirschey, 2021). Considering those large institutions, such as hedge funds, fit the impressions of fast traders best,[4] it is important to examine the effect of speed competition on price informativeness with large strategic traders.

To our best knowledge, this is the first paper to consider the *strategic* behaviour of large informed traders rather than small competitive traders in the context of speed acquisition. It designs a framework of strategic trading with two trading rounds that works *à la* Kyle (1985) to address these issues. Strategic insiders can invest in speed technology to trade before the information is capitalized into prices. In a benchmark model, we examine how speed competition affects price discovery and informativeness when large traders consider their price impact and strategically shade their trading aggressiveness. We find that the overall price informativeness with large strategic traders generates a hump shape to speed competition. This result goes opposite to a U-shaped result in Huang and Yueshen (2021) with a competitive

[1] For instance, Malikov and Pasquariello (2024) study the effects of quantitative investing on financial market quality via a model of strategic speculation with two informed traders. Discretionary investing is defined as fully strategic and quantitative investing as partially or fully myopic via its reliance on a backtested trading strategy.

[2] Such speed technology is often referred to as high-frequency trading (HFT) technology, but more broadly to how algorithms automate the whole trading process, from information processing (live streaming of news and data like Bloomberg) to risk management and compliance (Bouvard and Lee, 2016).

[3] For instance, by separating competitive investors' speed from information acquisition decision, Huang and Yueshen (2021) found that price discovery is a convex function in the population size, which renders the concentration of speed choice (in either early or late period), benefiting overall price informativeness.

[4] For example, hedge funds' high–frequency speed advantage arises from their investment in trading technology. The U.S. Securities and Exchanges Commission (SEC) defines high-frequency traders as "proprietary trading firms", "proprietary trading desks of a multi-service broker-dealer", or "hedge funds" (Section IV.B, SEC, 2010).

framework, showing that strategic trading of large traders can affect price informativeness differently from competitive trading of small traders.

The core of the model is how *strategic* trading reshapes the '*temporal fragmentation effect*' of the speed technology described by Huang and Yueshen (2021). The temporal fragmentation effect splits the trading process temporally into two parts, i.e., an early fragment with fast speculators and a late fragment with slow ones. Internalizing the price impact of trades, an informed trader cautiously shades trading aggressiveness on information to resolve a fraction of the *pre-trading uncertainty* of the fundamental, which is proportional to the number of traders in each trading round. As an inherent feature of the trading protocol à la Kyle (1985), the price discovery in the late period depends on not only the number of slow traders but also the amount of fundamental uncertainty resolved earlier by fast traders. This is different from Huang and Yueshen (2021) in which price discovery is independent across two periods. More explicitly, speed acquisition and strategic trading of large traders lead to two channels for price discovery.

First, embedding with the strategic behavior, fast traders not only contribute to price discovery in the early period by impounding information through their trades but also continue to impact price discovery in the late period as they help resolve the pre-trading fundamental uncertainty for slow traders. Therefore, fast trading is considered highly efficient in contributing to the overall price informativeness as the information obtained from fast trading can be amplified by slow traders. Hence, compared to the competitive models with temporal fragmentation where fast trading has no impact on late-period price discovery, the concertation of slow traders ultimately harms the overall price informativeness.

Second, to fully comprehend the information brought by fast trading, it is crucial to have the presence of slow traders alongside fast traders. The level of late-period price discovery is determined by the collective influence of slow traders and intermediate price informativeness. Slow trading acts as a catalyst for intermediate price informativeness, amplifying the effects of fast trading. When fast trading is concentrated, despite having a relatively high level of intermediate price informativeness, the late-period price discovery and the overall price informativeness suffer from the absence of slow traders who can reap the benefits of fast trading.

Speed competition, described by more traders switching from slow to fast, shifts the concentration of traders' speed choices. It generates two competing forces for price discovery in the late period; a positive effect, named as '*uncertainty reduction effect*', due to less pre-trading uncertainty from higher intermediate price informativeness, and a negative impact, called *'crowding-out effect*', due to decreasing number of traders staying late. Intensified speed competition, on the one hand, makes it easier for the slow to resolve uncertainty. On the other hand, it renders fewer speculators to contribute to price discovery in the late period. Therefore, the overall price informativeness generates a hump shape to speed competition.

To further examine how information dissemination and high-frequency trading affect the relationship between speed competition and overall price informativeness, we consider two extensions of the benchmark model. The first extension considers the level of information dissemination and speed competition. Not only fast trading, but technology also facilitates

information dissemination through social media. They bring new challenges to understanding the role of information aggregation in financial markets in this diverse and fast age.[5] With a speed acquisition, more precise payoff information is gradually revealed in the early and late periods among traders. Therefore, fast traders trade in the earlier period with less precise information while more precise information is revealed to slow traders in the late period. This information dissemination, measured by an incremental precision of information over time, captures sequential information arrivals and the time cost of information as in Kendall (2018) and Dugast and Foucault (2018).

We show that the hump-shaped relationship between these two variables remains robust when information dissemination is low. However, as the rate of information dissemination increases, the overall price informativeness decreases monotonically with the number of fast traders. More explicitly, due to the strategic trading behavior, the amount of price discovery is determined by the *pre-trade price informativeness* scaled by the number of fast and slow traders, but with different scales compared to the benchmark case. In addition to the amount of fundamental uncertainty resolved earlier from fast traders, price discovery in the late period further depends on information dissemination, in addition to the number of slow traders. Slow trading not only acts as a catalyst for intermediate price informativeness and amplifies the effects of fast trading but also impounds more precise information into the market due to sequential information arrivals, generating a new '*sequential information effect*'. Concentrated fast trading contributes positively to pre-trading uncertainty but harms late-period price discovery due to the absence of slow traders who can reap the benefits of fast trading and impound new information. Therefore, when the information diffusion is fast, the negative effect dominates the equilibrium and harms overall price informativeness.

The second extension renders fast traders trade at both early and late periods and examines how *high-frequency trading* affects the relationship between speed competition and overall price informativeness. The price informativeness increases in more intensified speed competition. Different from the benchmark case, the intensified speed competition does not reduce the number of slow traders resolving the long-run price discovery due to all traders trading at the late period. At the same time, as the speed competition becomes more intensified, more traders concentrate on the early period, which leads to less pre-trading uncertainty for the late period. Therefore, speed competition benefits price informativeness.

[5] Social media (such as tweets, newswires, and company reports, etc.) are different from traditional channels (such as newspapers or television). They have fundamentally changed how information is spread among traders and disseminated over time. Take Twitter as an example; people receive short electronic messages known as "*tweets*" through the Twitter accounts they follow, and they can re-transmit the messages by retweeting. This does not affect the initial information traders observe but incredibly influences transmission speed. Shiller (2015) also mentions, "*We were witnessing another explosion of technological innovations that facilitate interpersonal communication, consisting of e-mail and chat rooms, and after 2000 these led to social media… These new and effective media for interactive (if not face-to-face) communication may have the effect of expanding yet again the interpersonal contagion of ideas. They may have allowed enthusiasm for the market to spread much more widely than it would otherwise have. Certainly, we are still learning how to regulate the use of these new media in the public interest*".

1.1 *Related Literature*

This paper bridges two streams of literature: informed trading with (i) large strategic traders and (ii) varying speeds, see Table 1 for a summary of speed characteristics in sequential trading models in the literature. Based on Kyle (1985) and Kyle (1989), an extensive literature argues that private information is transmitted into asset prices through the trading of informed traders.[6] This literature largely abstracts from how the endogenous speed hierarchy, which temporarily fragments the progress of information aggregation, influences price informativeness. To our best knowledge, this is the *first* paper to consider the *strategic* behaviour of large informed traders rather than small competitive traders in the context of speed acquisition.

[Insert Table 1 here]

The concept of speed in this paper draws from the literature on sequential trading rounds. Hirshleifer et al. (1994) develop a model with two types of informed traders: '*early informed*' traders, who acquire information about fundamentals when the market opens at date 1, and '*late informed*' traders, who receive the same information only at date 2. Both types participate in trading across two periods. Similarly, Banerjee et al. (2018) distinguish between long-lived traders, who trade in both periods, and short-lived traders, who trade only in the later period while interpreting public signals differently.

Like these models, our framework also features sequential trading rounds, but with a key difference: fast traders exclusively trade in the early period, while slow traders only trade in the late period. Therefore, our approach is more closely aligned with Froot et al. (1992), who construct a model where half of the traders have their orders executed at date 1 and the other half at date 2, though traders are uncertain about the timing of execution when submitting their orders. In contrast, our model assumes that slow traders submit their orders after the early period, allowing for explicit interaction between early and late traders. Additionally, whereas Froot et al. (1992) treat the trading sequence as random, speed in our model is endogenously determined and a key variable of interest for speed acquisition.

Our model also contributes to the literature on endogenous speed acquisition. Holden and Subrahmanyam (1996), Dugast and Foucault (2018), and Kendall (2018) explore the endogenous decision to trade early or late, examining the trade-offs between different types of information signals. Holden and Subrahmanyam (1996) model short-horizon investors, who acquire short-term signals and close their positions in the next period, versus long-horizon investors, who acquire long-term signals and trade across both periods. Dugast and Foucault (2018) study information sellers who provide '*raw*' (unfiltered) and '*processed*' (filtered) signals about asset payoffs, with investors who buy raw signals trading earlier, while those who purchase processed signals trade later, reflecting the time needed for noise filtering. Kendall

[6] For instance, Banerjee and Breon-Drish (2019) consider a strategic trading environment to study dynamic information acquisition and entry that are detectable by the market maker. Xiong and Yang (2023) develop a theory to study the observability of strategic investors' information acquisition in financial markets.

(2018) introduces a choice between obtaining a weak, private signal early ('*rushing*') or spending time to obtain a stronger signal later ('*waiting*').

However, in these prior works, speed arises as a by-product of different types of information. Huang and Yueshen (2021) advance the literature by analyzing the acquisition of speed independently from information. In a competitive setting, Huang and Yueshen (2021) show that speed technology leads to a U-shaped relationship between speed competition and price informativeness. This is because the price discoveries have increasing returns to scale and are not interdependent, which makes them be convex functions in the population size.

Unlike Huang and Yueshen (2021), our model incorporates strategic traders. We show that under strategic trading, the relationship between price informativeness and speed cost is hump-shaped. Considering the price impact, traders strategically choose their trading aggressiveness to resolve a fraction of the *pre-trading uncertainty* of fundamentals, which is *proportional to the number of traders* in each trading round. This key mechanism inherits from strategic trading behaviour as in Kyle (1985) and Kyle (1989). Thus, strategic trading introduces a different dynamic in how speed competition can affect information aggregation.

The paper is organized as follows. Section 2 develops a model of speed competition and information diffusion with strategic insiders and characterizes market equilibrium. Section 3 discusses the speed competition and provides price discovery and informativeness implications. By incorporating information diffusion and high-frequency trading, Section 4 examines how information diffusion affects the nature of speed competition and price informativeness and Section 5 examines how high-frequency traders, described by fast traders trading at both periods, influence strategic trading and price informativeness. Section 6 concludes. All the proofs and additional discussions are presented in the Appendix.

2. The Model

We introduce a benchmark model of *speed competition* with *strategic* traders in this section to examine how fast trading cost affects speed acquisition, strategic trading of fast and slow traders, price discovery, and price informativeness in Section 2. Embedded with speed acquisition decision, this paper designs a framework of informed trading with two trading rounds that works *à la* Kyle (1985). In the following Sections 4 and 5, we extend the benchmark model to further examine how information dissemination and high-frequency trading affect the relationship between speed competition and overall price informativeness.

2.1. Financial Market with Speed Competition

Assets and Traders: Consider a financial market with a risky asset, a risk-free numéraire, and N strategic traders who are risk-neutral and fully informed. There are two trading rounds; $t = 1,2$, and the trading protocol mirrors Kyle (1985). At the end of the second trading round, each unit of the risky asset has a payoff $\tilde{V} \sim N(V_0, \sigma^2)$ measured by the numéraire. Without loss of

generality, V_0 is normalized to zero. As in the standard Kyle models, in each trading period, there is a liquidity trader who trades $\tilde{z}_t \sim N(0, \sigma_z^2)$ shares of the risky asset randomly. Also, a competitive market-maker absorbs the net trading flow and sets market prices p_t for an expected zero profit.

Speed Hierarchy: Speed technology can affect speculators' trading time (see "Timeline" in Figure 1 (a)). Based on the speed technology investment decision at $t = 0$, the N fully informed traders are grouped into two types: F *fast traders* who pay a speed cost C (> 0) to trade in the earlier period at time $t = 1$; and the remaining $S = N - F$ *slow traders* who trade in the later period at time $t = 2$. To capture trader's speed advantage to avoid intensified competition, we assume that the fast traders only trade in the earlier period, and no traders can trade in both periods. In this sense, fast traders are just fast but not high frequency. We assume that traders' speed acquisition decisions are overt, that is each trader knows other traders' speed choice.[7] We examine the effect of high-frequency trading in Section 5.

Speed Acquisition: The exogenous cost parameter C can be used to measure the speed technology level; the smaller, the cheaper (more advanced) the technology. Firstly, the speed acquisition decision is made at time $t = 0$ before traders observe their information. Trading technologies are proprietary and their emergence takes years to complete. With the ubiquity of computerized algorithmic and data abundance, speed hierarchy plays a vital role in financial markets. However, the infrastructure of fast trading technology to reduce order execution latencies is not in demand; they are made before traders observe the information. Secondly, fast traders have sufficient knowledge and experience to allow them to distinguish between fast and slow traders in the market before trading.[8] Therefore, we assume that traders know exactly the number of fast and slow traders in the market when participating in the market.

The speed characteristic and acquisition decision considered in this paper are motivated by the literature that features sequential trading rounds, such as short-horizon versus long-horizon in Holden and Subrahmanyam (1996), raw versus processed in Dugast and Foucault (2018), quick and imprecise versus slow and precise signals in Kendall (2018), and trade earlier versus stay slow in Huang and Yueshen (2021). In this literature, traders are small in the sense that each trader is a price-taker and does not have a price impact. However, institutions are often large traders. They can trade strategically and have price impacts, particularly with speed or information advantages. Therefore, their trading behavior and market impact are different when large traders trade strategically compared to small competitive traders. The strategic

[7] Xiong and Yang (2023) study the observability of investors' information-acquisition activities in financial markets and assume that there exists a communication technology that can potentially reveal investors' information-acquisition activities to the public. Under the overt benchmark setting, messages perfectly reveal investors' information precision so that their information-acquisition activities are completely observable. Under the secret benchmark setting, messages do not directly reveal any information regarding investors' information precision, and thus, their activities are completely unobservable. For simplicity, we treat the speed acquisition overt in our setup.

[8] While individual traders may not know the exact number of fast traders, they have access to market signals or proxies (e.g., order book data or observable trading speed) that allow them to infer the general distribution of fast and slow traders.

behaviour setting is particularly suitable for studying large players' trading in financial markets, such as large institutions.

2.2. Equilibrium Definition

A perfect Bayesian equilibrium of the trading game can be characterized by i) the equilibrium prices, $p_1^*(w_1)$, $p_2^*(w_1, w_2)$, over the two trading periods, ii) a strategy profile of the optimal demand of the fast traders, $x^{F,*}(V)$, and the slow traders, $x^{S,*}(V, p_1)$, conditional on different information they have received, and iii) the number of the fast traders, F^*, such that

$$\{p_1^*(w_1), p_2^*(w_1, w_2), x^{F,*}(V), x^{S,*}(V, p_1), F^*\}$$

satisfy

(i). profit maximization:

$$x^{F,*} = \underset{x^F}{argmax}\, E[x^F(V - p_1)|V]; \qquad x^{S,*} = \underset{x^S}{argmax}\, E[x^S(V - p_2)|V];$$

(ii). market efficiency: *p_1 and p_2 are determined according to weak-form-efficiency pricing rule*:

$$p_1 = E(V|w_1); \qquad p_2 = E(V|w_1, w_2);$$

(iii). speed acquisition: *Given the fast and slow traders' ex-ante expected profit*:

$$\pi_F = E\big[E[x^F(V - p_1)|V]\big]; \qquad \pi_S = E\big[E[x^S(V - p_2)|V, p_1]\big]; \tag{1}$$

and speed cost C, the equilibrium number of fast traders F^* is determined by

$$\pi_F(F^*, N - F^*) - C \geq \pi_S(F^* - 1, N - (F^* - 1)); \tag{2}$$

$$\pi_S(F^*, N - F^*) \geq \pi_F(F^* + 1, N - (F^* + 1)) - C$$

for $F^* \geq 1$ or $F^* = 0$ when $\pi_F - \pi_S < C$.

Equation (2) indicates that, in equilibrium, no trader wants to switch from fast (slow) to slow (fast). The traders in our model are large; hence, they consider not only the price impact as Kyle (1985) but also the impacts when choosing their speed. This means that, when switching from slow to fast (from fast to slow), they recognize that the number of fast traders increases (decreases).

We are interested in linear and "type-symmetric" perfect Bayesian equilibria in which equilibrium strategies are linear, and traders of the same type and information make the same decision and take the same action. This is standard in the traditional models of Kyle (1985). Specifically, a *linear equilibrium* is defined as a perfect Bayesian equilibrium in which there exist constants $(\beta_F, \beta_S, \gamma_S, \lambda_1, \lambda_2) \in \mathbb{R}^5$ such that

$$x^F = \beta_F V; \qquad x^S = \beta_S V - \gamma_S p_1;$$

$$p_1 = \lambda_1 w_1 \quad \text{with} \quad w_1 = Fx^F + z_1;$$

$$p_2 = p_1 + \lambda_2 w_2 \quad \text{with} \quad w_2 = Sx^S + z_2.$$

We explore the equilibrium properties in Section 3.

3. Equilibrium Properties

This section focuses on the impact of speed competition, reflected by the number of fast traders, on price informativeness. The number of fast and slow traders is exogenously given first and then endogenized by considering the cost of fast trading.

3.1. The Financial Market Equilibrium

We first study traders' optimal demand schedules for a given speed acquisition decision (i.e., for given numbers of fast and slow traders). All rational traders receive complete information about the fundamentals, only differ in their trading speed. The standard conjecture-and-verify analysis, as in Kyle (1985), yields the following results (we provide all the proofs in the Appendix).

LEMMA 1: *Given the number of fast traders F, there exists a linear Bayesian equilibrium with*

$$\beta_F = \frac{1}{\sqrt{F}}\frac{\sigma_z}{\sigma}; \qquad \beta_S = \gamma_S = \sqrt{\frac{F+1}{N-F}}\frac{\sigma_z}{\sigma}; \qquad \lambda_1 = \frac{\sqrt{F}}{F+1}\frac{\sigma}{\sigma_z}; \qquad \lambda_2 = \sqrt{\frac{N-F}{F+1}}\frac{1}{N-F+1}\frac{\sigma}{\sigma_z}.$$

The intermediate and overall price informativeness $\xi_t = [VAR(V|w_{\forall\tau\le t})]^{-1}$ are described by

$$\xi_1 = (F+1)\sigma^{-2}; \qquad \xi_2 = (F+1)(N-F+1)\sigma^{-2}.$$

Accordingly, the early and late price discovery $\Delta_t = \xi_t - \xi_{t-1}$ with $\xi_0 = \sigma^{-2}$ are described by

$$\Delta_1 = F\sigma^{-2}; \qquad \Delta_2 = (F+1)(N-F)\sigma^{-2}.$$

The ex-ante expected profits of fast and slow traders are given by, respectively,

$$\pi_F = \frac{1}{(F+1)\sqrt{F}}\sigma\sigma_z; \qquad \pi_S = \frac{1}{(N-F+1)\sqrt{N-F}\sqrt{F+1}}\sigma\sigma_z.$$

□

We focus on the impact of speed competition on price informativeness with strategic large traders. As in Huang and Yueshen (2021), fast trading temporally fragments price discovery. Therefore, the overall price informativeness fragments into two parts: the price discovery due

to the fast trading in the earlier period Δ_1 and the price discovery due to the slow trading in the later period Δ_2. We examine how the split of the total signal precision across the two trading periods affects the intermediate price informativeness ξ_1 and the overall price informativeness ξ_2.

3.2. Speed Competition with Strategic Trading

To begin with, Proposition 1 summarizes how intensified speed competition, described by the increasing number of fast traders, impacts intermediate and overall price informativeness.

PROPOSITION 1: *As speed competition becomes more intensified (i.e., the number of fast traders F increases),*

(i) the intermediate price informativeness ξ_1 monotonically improves, i.e., $\partial\xi_1/\partial F > 0$;

(ii) the overall price informativeness ξ_2 improves initially but reduces eventually, exhibiting a hump-shaped relationship with a maximum at $\bar{F} = N/2$, i.e., $\partial\xi_2/\partial F > 0$ (< 0) for $F < N/2 (> N/2)$.

□

Intuitively, Proposition 1 shows that, when the number of fast traders increases, the fast traders, as a group, trade more aggressively on fundamental information. This is described by an increasing aggregate trading intensity, i.e., $F\beta_F = \sqrt{F}\sigma_z/\sigma$, in the number of fast traders.[9] Hence, the price discovery due to the fast trading Δ_1 and the intermediate price informativeness ξ_1 increases in the number of fast traders F.

However, this speed competition has a non-monotonic effect on the overall price discovery due to the slow trading Δ_2 and the price informativeness ξ_2 as shown in Figure 2 (a). Even though the speed competition in our model shares similar features about the *"temporal fragmentation effect"* of speed technology in Huang and Yueshen (2021), Proposition 1 (ii) and Figure 2 (a) indicate that the overall price informativeness ξ_2 is *hump-shaped* in speed competition F. This result is opposite to Proposition 3.3 in Huang and Yueshen (2021). With small competitive traders, they show that, as the cost to be the fast decreases (i.e., as the number of fast traders increases), the price informativeness decreases initially (for the sufficiently high cost of speed acquisition) but increases eventually (for the sufficiently low cost of speed acquisition), exhibiting a *U-shaped* relationship to the cost of the speed. Therefore, strategic trading in our model plays a very different role in speed competition

[9] In Lemma 1, coefficients β_F and β_S measure the trading sensitivity of individual fast and slow traders on their information. They indicate that, with more fast (slow) traders, the individual fast (slow) traders trade less aggressively due to intensified competition, as in the standard Kyle (1985). Furthermore, the increasing fast traders impound more information into the equilibrium price in the earlier period. As a result, the market maker is better informed, leading to a more liquid price in the later period, indicated by lower λ_2. Therefore, the slow traders' trading intensity increases with the number of fast traders.

compared to competitive traders as in Kendall (2018), Dugast and Foucault (2018), and Huang and Yueshen (2021).

[Insert Figure 2 here]

We next examine why strategic trading leads to different results from competitive trading. We address this question from three different aspects: i) the constant returns to scale in price discovery, ii) the dependence of the late period price discovery on the number of fast traders, and iii) the split of the total signal precision across two trading periods.

***The constant returns to scale in price discovery*:** Compared to the competitive models (where every trader is atomlessly small as in Kendall, 2018; Dugast and Foucault, 2018; and Huang and Yueshen, 2021), the traders in our model are large and behave strategically. Considering the price impact, they strategically choose their trading aggressiveness to resolve a fraction of the *pre-trading uncertainty* of fundamentals, which is *proportional to the number of traders* in each trading round. Hence, the returns to scale of the pre-trading uncertainty and the number of traders are constant in the price discovery. Specifically, the amount of the price discovery contributed by the fast and slow traders are, respectively, given by

$$\Delta_1 = F\xi_0; \qquad \Delta_2 = (N-F)\xi_1, \tag{3}$$

where $\xi_0 = \sigma^{-2}$ is the initial price informativeness. Equation (3) shows that the amount of price discovery in the short-term (long-term) is simply the pre-trade price informativeness scaled by the number of fast (slow) traders. This result is standard in Kyle's one-period strategic trading model. However, this is very different from the competitive trader model, such as Huang and Yueshen (2021), where the price discovery in period t is proportional to the fraction of the traders trading at time t ($\Phi_t^2\sigma_z^{-2}$).

***The late period price discovery depends on the number of fast traders*:** The constant returns to scale of the pre-trading uncertainty in price discovery leads to the late period price discovery Δ_2 depending on the number of fast traders that resolve part of fundamental uncertainty earlier. Considering the intensified speed competition as described by more fast traders, it generates a trade-off about the late period price discovery and hence the overall price informativeness, depicted by

$$\frac{d\Delta_2}{dF} = \underbrace{(N-F)\frac{d(\xi_0+\Delta_1)}{dF}}_{uncertainty\ reduction\ effect} + \underbrace{(\xi_0+\Delta_1)\frac{dN-F}{dF}}_{crowding-out\ effect}. \tag{4}$$

On the one hand, the more uncertainty is resolved by the fast in the early period, the easier the slow traders to resolve uncertainty (all else equal) in the late period, generating an '*uncertainty reduction effect'*.[10] On the other hand, given the pre-trading uncertainty, more intensified

[10] This uncertainty reduction effect is also one of the driving forces in Goldstein and Yang (2015) and Xiong and Yang (2021). In Goldstein and Yang (2015), a higher level of trading intensity of one group of traders implies that there is more information about what they know in the price, which reduces the residual uncertainty for other type of informed traders face when they trade and hence induces them to trade more aggressively. In Xiong and Yang (2021), speculators face uncertain demand shocks in the spot price when trading futures contracts, and releasing information about these shocks reduces their uncertainty. Being risk averse, speculators then trade more

speed competition (or fast traders) crowds out slow traders and renders fewer speculators to contribute to price discovery in the late period, producing a '*crowding-out effect'*.

Therefore, the impact of a marginal change in fast traders on the price discovery in the later period depends on the initial number of fast traders F. When F is close to zero, an increase in F prompts a small number of traders to move from slow to fast. However, this does not dramatically reduce the number of slow traders. Also, the more informative early equilibrium price benefits all the slow traders (whose population is relatively large). Hence, the positive contribution of the slow traders to the price discovery from less pre-trading uncertainty becomes more significant compared to the potentially negative effect of fewer slow traders (when the number of fast traders increases). Accordingly, the uncertainty reduction effect dominates the crowding-out effect and the price discovery improves. However, the reverse holds when the number of fast traders is high, close to N. In this case, the crowding-out effect dominates the information effect, reducing price discovery.

***Split of the total signal precision across two trading periods*:** Essentially, the speed technology temporally fragments the speculator's participation in the market, shifting the concentration of the price discovery process. The novel '*uncertainty reduction effect'*, combining the splits of the total signal precision across the two trading periods, differentiates our study from the competitive models such as Huang and Yueshen (2021).

Note that, with competitive traders, the late period price discovery is proportional to the amount of *noise trading precision* σ_z^2, the standard outcome from competitive models à la Hellwig (1980) with market clearing; or à la Vives (1995) with competitive market makers. This means that the amount of uncertainty resolved in the early period has no impact on the late-period price discovery. Hence, the overall price informativeness can be described by

$$\xi_2 = \xi_0 + \Delta_1 + \mathcal{G}(S),$$

where $\mathcal{G}(S) = S^2/(\gamma\sigma_z)^2$ and $\Delta_1 = \tau_U F^2/(\gamma\sigma_z)^2$. The key to the U-shaped relationship in Huang and Yueshen (2021) is as follows. Since the price discoveries have increasing returns to scale and are not interdependent, price discovery is a convex function in the population size. Namely, each marginal informed investor's trading resolves increasingly more uncertainty (from noise trading). Therefore, a more concentrated distribution of the same type of trader benefits price informativeness. However, in our setup, the amount of uncertainty resolved in the early period affects the late period price discovery,

$$\xi_2 = \xi_0 + \Delta_1 + \mathcal{H}(\Delta_1, S),$$

where $\mathcal{H}(\Delta_1, S) = (\sigma^{-2} + \Delta_1)S$. We can rewrite the overall price informativeness in Lemma 1 as

$$\xi_2 = (F+1)(N-F+1)\sigma^{-2} = \sigma^{-2} + F\sigma^{-2} + (F+1)(N-F)\sigma^{-2}.$$

Therefore, the novel '*uncertainty reduction effect'* leads the late-period price discovery, as well as the overall price informativeness, to be the summation of the convex functions. Hence, as more traders choose to be fast, the concentration of the price discovery process shifts, and the overall price informativeness increases in the product of the number of fast traders. The hump

aggressively on their own private information, injecting more information into the futures price and enabling firms to make better production decisions.

shape relationship in Proposition 1 (ii) is hence an inherent feature of the trading protocol à la Kyle (1985), a result of the strategic behavior of large informed traders. We summarize the above discussion about the mechanism in Figure 3.

[Insert Figure 3 here]

3.3. Advanced Speed Technology with Strategic Trading

Considering the costs of fast trading, we endogenize the number of fast traders in Lemma 2. Since traders in our model are large, we need to pay more attention to their impacts—not only the price impacts in Kyle (1985) but also the impacts when choosing their speed, as indicated by Equation (2).

LEMMA 2: *Given speed cost* C*, let* X *be the unique solution of function* $\mathcal{G}(X) = 0, where$

$$\mathcal{G}(X) = \frac{1}{(X+1)\sqrt{X}}\sigma\sigma_z - \frac{1}{(N-X+1)\sqrt{N-X}\sqrt{X+1}}\sigma\sigma_z - C.$$

The equilibrium number of fast traders then depends on the value of X*,*

(i) when

$$X \geq \bar{X} = \frac{5N - 1 - \sqrt{(3N+1)^2 + 16(N+1)}}{8},$$

the equilibrium population of fast traders is determined by

$$F^* = \begin{cases} \lceil X \rceil, & \frac{1}{(\lceil X \rceil + 1)\sqrt{\lceil X \rceil}}\sigma\sigma_z \geq \frac{1}{(N - \lfloor X \rfloor)\sqrt{N - \lfloor X \rfloor + 1}\sqrt{\lfloor X \rfloor}}\sigma\sigma_z + C; \\ \lfloor X \rfloor, & otherwise. \end{cases}$$

(ii) when $X < \bar{X}$*, the equilibrium population of fast traders is determined by*

$$F^* = \begin{cases} \lfloor X \rfloor, & \frac{1}{(\lfloor X \rfloor + 1)\sqrt{\lfloor X \rfloor}}\sigma\sigma_z \geq \frac{1}{(N - \lfloor X \rfloor)\sqrt{N - \lfloor X \rfloor + 1}\sqrt{\lfloor X \rfloor}}\sigma\sigma_z + C; \\ \lfloor X \rfloor - 1, & otherwise. \end{cases}$$

where $\lfloor \cdot \rfloor$ *and.* $\lceil \cdot \rceil$ *represent the floor and ceil functions, respectively.*

□

Lemma 2 depicts the endogenous equilibrium number of fast traders from the integral problem. Note that switching to the break-even condition of no integral problem does not qualitatively change our results. Therefore, we simplify the integral problem and consider the break-even condition for $F \in R^+$ in the rest of the paper. Accordingly, we can relate speed technology to price informativeness in Proposition 2.

PROPOSITION 2: *As the speed technology becomes more advanced (i.e., the speed cost C decreases),*

(i) the intermediate price informativeness ξ_1 monotonically improves, i.e., $\partial\xi_1/\partial C < 0$;

(ii) the overall price informativeness ξ_2 improves initially but reduces eventually, exhibiting a hump-shape relationship with a maximum at $\bar{C}$, i.e., $\partial\xi_2/\partial C > 0$ for $0 \le C < \bar{C}$; $\partial\xi_2/\partial C < 0$ for $\bar{C} \le C < \hat{C}$, where

$$\bar{C} = \frac{2\sqrt{2N+4}-4}{(N+2)\sqrt{N}\sqrt{N+2}}\sigma\sigma_z, \qquad \hat{C} = \frac{N\sqrt{N-1}-\sqrt{2}}{2N\sqrt{N-1}}\sigma\sigma_z.$$

□

We numerically illustrate Proposition 2 in Figure 2 (c) – (d). As speed technology becomes more advanced, the number of fast traders increases as in Figure 2 (c). Based on Proposition 2, there exists a hump-shaped relationship between the overall price informativeness and the cost of the speed acquisition as in Figure 2 (d). While the model is stylized, the temporal fragmentation effect from large and strategic traders is a robust feature of the speed technology. This temporal fragmentation negatively impacts the competitive model, for instance, Huang and Yueshen (2021), as the diversification of speculators' speed choice reduces price informativeness. In contrast, it shows a '*good*' fragmentation impact in our strategic model as the diversification of speculators' speed choice benefits price informativeness.

3.4. Empirical Implications

The advancement of speed technology has become one of the most salient developments in recent securities trading. This trend has driven a rapid rise in machine-driven trading, particularly characterized by speed. Empirically, extensive discussions have addressed factors that could influence stock market price informativeness. This paper outlines several testable empirical implications.

Prediction 1. (Speed competition and price informativeness). i) *Short-run price informativeness increases with speed competition*; ii) *There exists a hump-shaped relationship between speed competition and long-run price informativeness.*

□

Existing empirical studies appear to provide indirect support for these predictions. For example, short-term (intraday) measures of price efficiency (e.g., order flow predictability, serial correlation) seem to improve (Brogaard et al., 2014; Hirschey, 2018), while long-term price informativeness seems negatively affected (e.g., Weller, 2018). To formally test these theoretical predictions, we would need clear measures of both price informativeness and speed

competition. In the following, we suggest potential databases and outline a proposed empirical procedure.

Sample Construction: The main sample could include the U.S.-listed companies trading on the NYSE, NASDAQ, and AMEX, with firm-level financial statement data drawn from Compustat, supplemented by intangible capital estimates from Peters and Taylor (2017) in WRDS. Stock price information could be sourced from CRSP. Since this study focuses on strategic traders, metrics of speed competition might be constructed by extracting institutional holdings data from Thomson Reuters 13F and gathering fund strategies from the SEC's EDGAR database. Using these data, variables such as fast and slow institutional ownership concentration could be created for each firm-quarter.

For firm-level empirical tests, we might select the most recent ownership variables relative to each firm's fiscal year-end and then merge them with firm-level financial data. Sampling criteria would exclude firms with a stock price below $1 or a market cap below $500 million, as well as firms in the financial industry and those with fewer than four years of continuous accounting data. We could further require that sample firms have at least one institutional investor. For metrics that rely on firm-level concentration, requiring a minimum of five active institutional investors could prevent extreme values. Unless otherwise specified, these criteria would apply consistently throughout all the proposed analyses.

Measures of Speed Competition: Classifying institutions as fast or slow investors could follow Abis (2022) machine learning-based classification. This method uses data from the "Principal Investment Strategies" (PIS) section of mutual fund prospectuses, part of the N-1A filing since December 1999.[11] In this proposed procedure, a random sample of PIS sections would be reviewed by expert classifiers, with their classifications used to train a machine learning algorithm—a random forest—to categorize institutions as fast or slow. This algorithm is transparent and interpretable, providing a ranked list of features (e.g., keywords and combinations) that inform classification.

Firm-level fast institutional ownership could be calculated as

$$sc_{i,t} = \frac{\sum_{j=1}^{N_i} S_{i,j,t}}{\sum_{j=1}^{M_i} S_{i,j,t}}, \tag{5}$$

where $S_{i,j,t}$ denotes the equity shares of stock i owned by institution j at time t and N_i is the number of fast institutions holding stock i and M_i is the total number of institutions holding stock i.

Measures of Price Informativeness: The empirical literature provides various approaches to measuring price informativeness. Following Bai et al. (2016), we propose a metric for price informativeness, denoted as *FPE* (forecasting price efficiency), to capture

[11] Funds must "*Explain in general terms how the Fund's adviser decides which securities to buy and sell*". This requirement was added after the 1998 amendment of mandatory disclosures; all funds were required to comply beginning in December 1999. Focusing on mandatory disclosures allows for more reliable information than analyzing other self-reported material.

information on future cash flows embedded in prices. We can conduct the following regression of the future earnings,

$$\frac{E_{i,t+k}}{A_{i,t}} = a_{t,k} + b_{t,k}\, log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_{t,k}\left(\frac{E_{i,t}}{A_{i,t}}\right) + Controls. \quad (6)$$

where k denotes the prediction horizons; $M_{i,t}/A_{i,t}$ denotes the market price of firm i in fiscal year t, computed as the market capitalization at the end of March after year t, scaled by total assets in year t; $E_{i,t+k}/A_{i,t}$ $(E_{i,t}/A_{i,t})$ denotes future (current) earnings, computed as cash flow in year $t+h$ (t) scaled by total assets in year t. The market-level FPE in year t at prediction horizon k is then calculated as the forecasting coeffcient $b_{t,k}$ in Equation (6) multiplied by $\sigma_t(log(M/A))$ the cross-sectional standard deviation of the scaled market price $log(M/A)$ in year t:

$$FPE_{t,k} = b_{t,k} \times \sigma_t\left(log\left(\frac{M_t}{A_t}\right)\right).$$

The predicted variation is a measure of FPE, the amount of information about future cash flows contained in prices, which is the most relevant for the role of stock prices in capital allocation.

Running cross-sectional regressions could yield a time series of FPE measures, which could be used to examine in relation to market-level fast institutional ownership. However, this cross-sectional approach may not be well-suited for firm-level analysis, as firm-level speed competition is typically panel data, while price informativeness is time-series data. Kacperczyk et al. (2021) address this limitation by proposing a pooled regression model, which can be used to investigate the effects of firm-level speed competition on price informativeness.

Speed Competition and Price Informativeness: Firm-level Evidence: To examine the hypothetical effect of firm-level speed competition on FPE, we could follow Kacperczyk et al. (2021) and adopt a pooled regression model of future earnings on speed competition based on annual firm-level data,

$$\frac{E_{i,t+k}}{A_{i,t}} = \alpha_k + \beta_{k,1} \times sc_{i,t} + \beta_{k,2} \times sc_{i,t}^2 + (\gamma_{k,0} + \gamma_{k,1} \times sc_{i,t} + \gamma_{k,2} \times sc_{i,t}^2) log\left(\frac{M_{i,t}}{A_{i,t}}\right) + \delta_t\left(\frac{E_{i,t}}{A_{i,t}}\right) + Controls, \quad (7)$$

where k denotes the prediction horizons, $sc_{i,t}$ denotes the speed competition, measured by Equation (5). Other variables are defined as in Equation (6).

The coefficients $\gamma_{k,1}$ and $\gamma_{k,2}$ are of interest to capture the sensitivity of future earnings to current prices, conditional on speed competitions. Based on Prediction 1, we would expect a positive coefficient $\gamma_{k,1}$ for short-run such as $k = 1$, and insignificant coefficient $\gamma_{k,2}$. For long-run such as $k = 3$, the coefficient $\gamma_{k,2}$ is negative.

We consider two further extensions in the following. The first extension considers information dissemination and highlights the relationship between the overall price informativeness and speed competition. The second extension renders fast traders trade at both early and late periods and examines how high-frequency trading affects the relationship between speed competition and overall price informativeness.

4. Information Diffusion

This section examines how information diffusion influences speed competition, strategic trading, and price informativeness. Extensive evidence shows that information diffusion influences investors' trading behavior in financial markets. Therefore, it becomes more critical to understand how information diffusion influences speed competition and whether faster information dissemination improves price discovery and price informativeness.

4.1 The Setup

Apart from speed competition in the benchmark model, the main difference between the model introduced in this section and a standard Kyle model is that fast and slow traders receive information from a diffusion process. Specifically, the fast traders trade on more noisy fundamental information $s_a = v + \varepsilon + \zeta$ with independent noise components $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ and $\zeta \sim N(0, \sigma_\zeta^2)$ in the early period, while the slow traders trade on less noisy fundamental information $s_b = v + \varepsilon$ in the late period. For given information precision of the fast traders, i.e., the total variance $\sigma_\varepsilon^2 + \sigma_\zeta^2 = \sigma_f^2$, we use the fractional variance σ_ζ^2 of information s_a to the total variance ($\sigma_\varepsilon^2 + \sigma^2$) of information s_b to represent the inter-temporal information dissemination:

$$\eta = \frac{\sigma_\zeta^2}{\sigma^2 + \sigma_\varepsilon^2} \in \left[0, \frac{\sigma_f^2}{\sigma^2}\right]. \tag{8}$$

With a higher value of η, the information diffuses faster over time. Therefore, slow traders become more informative.

The information diffusion structure introduced is closely related to sequential information arrivals and the time cost of information. Therefore, η can also be used to measure the incremental precision of information diffusion over time. This information diffusion is similar to Kendall (2018) and Dugast and Foucault (2018), who examine the time cost of homogeneous information so that the precision of traders' information increases with time. Therefore, information filters out the noise in data, but it takes time.

4.2 The Equilibrium

We analyze the equilibrium in two steps. Given their speed acquisition, the first is to study speculators' optimal demand schedules. The standard conjecture-and-verify analysis, as in Kyle (1985), yields the financial market equilibrium. Traders then independently make their optimal speed acquisition decisions at time $t = 0$ of investing in speed technologies to be fast or remaining as slow traders. The decision is characterized by a trade-off on the trader's expected payoffs; trading earlier can increase the information rent, but speed acquisition is costly. Therefore, a trader chooses to become fast only when his expected payoff from trading in the early period is sufficient to cover the cost of speed acquisition, compared to the slow

traders. In an interior equilibrium, after considering the cost of speed acquisition, a trader is indifferent between being fast by acquiring speed and being slow; otherwise, a corner equilibrium arises. The resulting equilibrium is characterized in Lemma 3.

LEMMA 3: *For given information precision of the fast trader, i.e.,* $\sigma_\varepsilon^2 + \sigma_\zeta^2 = \sigma_f^2$*, there exists a linear Bayesian equilibrium with*

$$\beta_F = \frac{1}{\sqrt{F}} \frac{\sigma_z}{\sqrt{\sigma^2 + \sigma_f^2}}; \qquad \beta_S = \varpi \sqrt{\frac{F+1}{S}} \frac{(\eta+1)\sigma^2}{\sigma^2 + \sigma_f^2} \frac{\sigma_z}{\sigma}; \qquad \gamma_S = \varpi \sqrt{\frac{F+1}{S}} \frac{\sigma_z}{\sigma};$$

$$\lambda_1 = \frac{\sqrt{F}}{F+1} \frac{\sigma^2}{\sqrt{\sigma^2 + \sigma_f^2}} \frac{1}{\sigma_z}; \qquad \lambda_2 = \frac{1}{\varpi} \sqrt{\frac{S}{F+1}} \frac{1}{S+1} \frac{\sigma}{\sigma_z}; \qquad \varpi = \frac{\sqrt{\sigma^2 + \sigma_f^2}}{\sqrt{(F+1)\eta + 1}} \frac{1}{\sigma}.$$

The intermediate and overall price informativeness $\xi_t = [VAR(V|w_{\forall \tau \le t})]^{-1}$ *can be described by, respectively,*

$$\xi_1 = \frac{F+1}{\sigma^2} \frac{\sigma^2 + \sigma_f^2}{\sigma^2 + (F+1)\sigma_f^2};$$

$$\xi_2 = \frac{(F+1)(S+1)}{\sigma^2} \frac{\sigma^2 + \sigma_f^2}{[1 - (F+1)S\eta]\sigma^2 + (F+1)(S+1)\sigma_f^2}.$$

The ex-ante expected profits of fast and slow traders are given by, respectively,

$$\pi_F = \left[\frac{1}{(F+1)\sqrt{F}} \frac{\sigma}{\sqrt{\sigma^2 + \sigma_f^2}} \right] \sigma \sigma_z; \qquad \pi_S = \left[\frac{1}{(S+1)\sqrt{S}} \sqrt{\eta + \frac{1}{F+1}} \frac{\sigma}{\sqrt{\sigma^2 + \sigma_f^2}} \right] \sigma \sigma_z.$$

The equilibrium number of fast traders is uniquely determined by $\pi_F - C = \pi_S$ *when* $\eta < \hat{\eta}$ *and* $C < \hat{C}(\eta)$*, otherwise (i.e.* $\eta \ge \hat{\eta}$ *or* $C \ge \hat{C}(\eta)$*), all traders stay slow, where*

$$\hat{\eta} = \frac{(N-1)N^2 - 2}{4}; \qquad \hat{C}(\eta) = \frac{\sigma^2 \sigma_z}{\sqrt{\sigma^2 + \sigma_f^2}} \left[\frac{1}{2} - \frac{1}{\sqrt{(N-1)N}} \sqrt{\eta + \frac{1}{2}} \right].$$

□

Note that Lemma 3 reduces to Lemma 1 when $\sigma_\varepsilon^2 = \sigma_\zeta^2 = 0$ (or $\eta = 0$). Lemma 3 shows that information diffusion affects the strategic trading of speculators and market maker. It also affects the price informativeness and ex-ante expected profits. Furthermore, the equilibrium fraction of fast traders depends on the cost (or the level of speed technology) and information diffusion. When investing in speed is too costly ($C > \hat{C}$), or information diffusion is too fast ($\eta \ge \hat{\eta}$), no trader acquires speed to trade fast. Therefore, traders choose to be fast only for

sufficiently lower cost of the speed acquisition ($C < \hat{C}$) and relatively slow information diffusion.[12]

4.3 Equilibrium Properties

We first examine how intensified speed competition, described by the increasing number of fast traders, affects price informativeness. Similar to the analysis in Section 2, considering price impact, large traders strategically choose their trading aggressiveness so that each trading round resolves a fraction of the *pre-trading uncertainty* of the fundamental, depending on the number of traders. Specifically, the amount of price discovery contributed by the fast and slow traders is now given by, respectively,

$$\Delta_1 = \frac{F\sigma^2}{\sigma^2 + (F+1)\sigma_f^2}\xi_0; \qquad \Delta_2 = \frac{[S + (F+1)S\eta]\sigma^2}{[1-(F+1)S\eta]\sigma^2 + (F+1)(S+1)\sigma_f^2}\xi_1, \tag{9}$$

where $\xi_0 = \sigma^{-2}$ is the initial price informativeness. Equation (9) mirrors the results in Equation (3) that the amount of price discovery in the short-term (long-term) is simply the pre-trade price informativeness scaled by the number of fast (slow) traders but with different scales. Proposition 3 presents how intensified speed competition affects price informativeness.

PROPOSITION 3: *As speed competition becomes more intensified (i.e., the number of fast traders F increases),*

(i) the intermediate price informativeness ξ_1 monotonically improves, i.e., $\partial\xi_1/\partial F > 0$;

(ii) when $\eta < N$, the overall price informativeness ξ_2 improves initially but reduces eventually, exhibiting a hump-shape relationship with a maximum at $\hat{F}$, i.e., $\partial\xi_2/\partial F > 0$ (< 0) for $F < \hat{F}(> \hat{F})$, where

$$\hat{F} = \frac{\sqrt{1 + (2+N)\eta} - (1+\eta)}{\eta}, \tag{10}$$

(iii) when $N \leq \eta < \hat{\eta}$, the overall price informativeness ξ_2 monotonically reduces, i.e., $\partial\xi_1/\partial F > 0$.

□

Considering the intensified speed competition as described by more fast traders, apart from the two competition forces in Equation (4), it generates another force, '*information effect*', on price discovery and the overall price informativeness in the late period when considering inter-temporal information diffusion is depicted by,

[12] The speed cost measures the ease of acquiring speed. A significant improvement in trading technology (say, investments in computational hardware and fast connection to exchange servers) can reduce the cost of fast trading and make it cheaper for traders.

$$\frac{d(\xi_2-\xi_1)}{dF}=\underbrace{\frac{[S+(F+1)S\eta]\sigma^2}{[1-(F+1)S\eta]\sigma^2+(F+1)(S+1)\sigma_f^2}\frac{d\xi_1}{dF}}_{uncertainty\ reduction\ effect}+\xi_1\frac{d}{dF}\left[\frac{[S+(F+1)S\eta]\sigma^2}{[1-(F+1)S\eta]\sigma^2+(F+1)(S+1)\sigma_f^2}\right] \tag{11}$$

$$=\underbrace{\frac{[S+(F+1)S\eta]\sigma^2}{[1-(F+1)S\eta]\sigma^2+(F+1)(S+1)\sigma_f^2}\frac{d\xi_1}{dF}}_{uncertainty\ reduction\ effect}+\underbrace{\xi_1\frac{d}{dF}\left[\frac{S\sigma^2}{\sigma^2+(F+1)(S+1)\sigma_f^2}\right]}_{crowding-out\ effect}$$

$$+\underbrace{\xi_1\frac{d}{dF}\left[\frac{[S+(F+1)S\eta]\sigma^2}{[1-(F+1)S\eta]\sigma^2+(F+1)(S+1)\sigma_f^2}-\frac{S\sigma^2}{\sigma^2+(F+1)(S+1)\sigma_f^2}\right]}_{information\ effect\ (-)}.$$

This shows that, on the one hand, the more uncertainty is resolved by the fast in the early period, the easier the slow traders to resolve uncertainty (all else equal) in the late period, corresponding to the *uncertainty reduction effect*. On the other hand, given the pre-trading uncertainty, more intensified speed competition (or fast traders) crowds out slow traders and renders fewer speculators to contribute to price discovery in the late period, producing the *crowding-out effect*. Furthermore, since slow traders observe more precise information from information diffusion, more traders switch from slow to fast. This further deteriorates price discovery since less new information is impounded into the price, named as '*information effect*'.

As illustrated in Figure 4 (a), Proposition 3 (ii) shows that, with relatively slow information diffusion, the results in Proposition 1 remain robust. Furthermore, Proposition 3 (iii) shows that, when the information diffusion is relatively fast $N \le \eta < \hat{\eta}$, the negative impact through the information effect becomes significant, which renders the overall price informativeness to decrease monotonically with the number of fast traders. This is illustrated in Figure 4(b). Mathematically, there exists a threshold number of fast traders $\hat{F}$ such that the overall price informativeness ξ_2 improves for $F < \hat{F}$ *(i.e.,* $\partial\xi_2/\partial F > 0$) but reduces *(i.e.,* $\partial\xi_2/\partial F < 0$) for $F > \hat{F}$. According to Equation (10), the threshold value $\hat{F}$ can become negative with faster information diffusion. Hence, our model provides a cautionary tale of the disruptive effects of how technological advancement and faster information diffusion, as seen recently, might hinder the price discovery function of financial markets.

[Insert Figure 4 here]

We further explore the equilibrium properties with advanced speed technology using comparative statics regarding speed technology, depending on the rate of information diffusion. We summarize the results in Proposition 4 and illustrate them in Figures 4 (c) and (d).

PROPOSITION 4: *As the speed technology becomes more advanced (i.e., the speed cost* C *decreases),*

(i) when $\eta < N$*, the overall price informativeness* ξ_2 *improves initially but reduces eventually, exhibiting a hump-shape relationship with a maximum at* $\bar{C}$*, i.e.,* $\partial\xi_2/\partial C > 0$ *for* $0 \le C < \bar{C}$; $\partial\xi_2/\partial C < 0$ *for* $\bar{C} \le C < \hat{C}$*, where*

$$\bar{C}(\eta)=\frac{\sigma^2\sigma_z}{\sqrt{\sigma^2+\sigma_f^2}}\left[\frac{1}{(X+1)\sqrt{X}}-\frac{1}{(N-X+1)\sqrt{N-X}}\sqrt{\eta+\frac{1}{X+1}}\right]\Bigg|_{X=\frac{\sqrt{1+(2+N)\eta}-(1+\eta)}{\eta}};$$

$$\hat{C}(\eta)=\frac{\sigma^2\sigma_z}{\sqrt{\sigma^2+\sigma_f^2}}\left[\frac{1}{2}-\frac{1}{\sqrt{(N-1)N}}\sqrt{\eta+\frac{1}{2}}\right].$$

(ii) when $N \leq \eta < \hat{\eta}$*, the overall price informativeness* ξ_2 *monotonically reduces.*

□

Note that when $\eta = 0$ and $\sigma_f^2 = 0$, the threshold values $\bar{C}$ and $\hat{C}$ in Proposition 4 are simplified to the corresponding threshold values in Proposition 2. As η increases, the threshold value $\bar{C}$ decreases, shifting the hump-shape leftwards as in Figures 4 (c) and (d). Therefore, Proposition 4 shows two important results. Firstly, Proposition 4 (i) shows that the hump-shaped relationship of the overall price informativeness to speed technology due to temporal fragmentation effect remains when information diffusion is low, though the pattern of the hump shape can shift, depending on information diffusion η. Secondly, Proposition 4 (ii) shows that the inter-temporal information diffusion dissemination η has a significant impact on the relation between speed competition and overall price informativeness. With faster information diffusion, though advanced technology harms the overall price informativeness, information diffusion itself benefits price discovery as comparing Figures 4 (c) and (d).

5. High-Frequency Trading

This section examines how speed competition influences strategic trading and price informativeness when fast traders are high-frequency traders, meaning that fast traders can trade in both periods. The standard conjecture-and-verify analysis, as in Yang and Zhu (2020), yields the financial market equilibrium. To characterize perfect Bayesian equilibrium, the market maker's strategy is pinned down by the weak efficiency rule. We conjecture the following symmetric linear strategies,

$$x^F = \beta_F s_a; \qquad x^S = \beta_S s_b - \gamma_S p_1; \tag{12}$$

$$p_1 = \lambda_1 w_1 \quad \text{with} \quad w_1 = Fx^F + z_1; \tag{13}$$

$$p_2 = p_1 + \lambda_2 w_2 \quad \text{with} \quad w_2 = Nx^S + z_2. \tag{14}$$

Market maker's decision: In period 1, the market maker observes the aggregate order flow w_1 and sets $p_1 = E(V|w_1)$. Using the conjectured trading strategies and the projection theorem, we can compute

$$\lambda_1 = \frac{Cov(V, w_1)}{Var(w_1)} = \frac{F\beta_F\sigma^2}{F^2\beta_F^2(\sigma^2+\sigma_f^2)+\sigma_z^2}. \tag{15}$$

Similarly, in period 2, the market maker observes $\{w_1, w_2\}$ and sets $p_2 = E(V|w_1, w_2)$, which implies that

$$\lambda_2 = \frac{Cov(V, w_2|w_1)}{Var(w_2|w_1)}.$$

With the conjectured trading strategies, we apply the projection theorem and obtain

$$\lambda_2 = \frac{(\eta+1)\left(\eta F^2\beta_F^2\left(\sigma^2+\sigma_f^2\right)+(\eta+1)\sigma_z^2\right)N\beta_S\sigma^2}{\left(\eta F^2\beta_F^2\left(\sigma^2+\sigma_f^2\right)+(\eta+1)\sigma_z^2\right)N^2\beta_S^2\left(\sigma^2+\sigma_f^2\right)+(\eta+1)^2\left(F^2\beta_F^2\left(\sigma^2+\sigma_f^2\right)+\sigma_z^2\right)\sigma_z^2}. \quad (16)$$

Slow traders' decision: *Let* $E[x^S(V-p_2)|s_b, p_1]$ be the trading profit of a slow trader from her trade x^S in period 2. The slow trader observes $\{s_b, p_1\}$ and chooses x^S to maximize $E[x^S(V-p_2)|s_b, p_1]$. Using the conjectured trading strategies (12) and the pricing function (14), we can compute the first-order condition (FOC) and obtain

$$x^S = \frac{E[V-p_1|s_b, p_1]}{2\lambda_2} - \frac{E[(N-1)x^S|s_b, p_1]}{2}. \quad (17)$$

Based on the conjectured trading strategies, we apply the projection theorem and show that both $E[V-p_1|s_b, p_1]$ and $E[(N-1)x^S|s_b, p_1]$ are linear functions of s_b and p_1. Inserting these linear functions into (17), we compare this expression with the conjectured strategy (12) to arrive at the following equation:

$$\beta_S = \frac{1}{\lambda_2(N+1)}\frac{(\eta+1)\sigma^2}{\sigma^2+\sigma_f^2}; \qquad \gamma_S = \frac{1}{\lambda_2(N+1)}.$$

Fast trader's decision: Fast traders can trade in both periods. We solve their problems by backward induction. In period 2, the fast traders behave just like the slow traders. In period 1, the fast traders observe s_a and choose x^F to maximize $E[x^F(V-p_1) + E[x^S(V-p_2)|V, p_1]|V]$, where the second-period profit is generated from the optimal trading strategy (17). Direct computation shows that

$$E[x^F(V-p_1) + E[x^S(V-p_2)|s_b, p_1]|s_a]$$
$$= x^F\left\{\frac{\sigma^2}{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}s_a - \lambda_1[x_1+(F-1)\alpha s_a]\right\} + \frac{1}{\lambda_2(N+1)^2}E\left[\left(\frac{\sigma^2}{\sigma^2+\sigma_\varepsilon^2}s_b - p_1\right)^2|s_a\right].$$

We assume that the fast traders play pure strategy. From the first-order condition (FOC), we have the expression of β_F

$$\beta_F = \frac{1-\frac{2\lambda_1}{\lambda_2(N+1)^2}}{(F+1)\lambda_1 - \frac{2\lambda_1^2 F}{\lambda_2(N+1)^2}}\frac{\sigma^2}{\sigma^2+\sigma_f^2}.$$

By focusing on the equilibrium without information diffusion, we have Lemma 4.

LEMMA 4: *When both fast and slow traders are fully informed, there exists a linear Bayesian equilibrium with*

$$\beta_F = \frac{\lambda_2(N+1)^2 - 2\lambda_1}{\lambda_1[\lambda_2(N+1)^2(F+1) - 2\lambda_1 F]}; \quad \beta_S = \gamma_S = \frac{1}{\lambda_2(N+1)}; \quad \lambda_1 = \frac{F\beta_F\sigma^2}{F^2{\beta_F}^2\sigma^2+\sigma_z^2}; \quad \lambda_2 = \frac{N\beta_S\sigma^2}{F^2{\beta_F}^2\sigma^2+N^2{\beta_S}^2\sigma^2+\sigma_z^2}.$$

The intermediate and overall price informativeness $\xi_t = [VAR(V|w_{\forall\tau\leq t})]^{-1}$ *can be described by, respectively,*

$$\xi_1 = \frac{(F\beta_F)^2\sigma^2 + \sigma_z^2}{\sigma_z^2}\xi_0; \qquad \xi_2 = \frac{(F\beta_F)^2\sigma^2 + (N\beta_S)^2\sigma^2 + \sigma_z^2}{(F\beta_F)^2\sigma^2 + \sigma_z^2}\xi_1.$$

Accordingly, the early and late price discovery $\Delta_t = \xi_t - \xi_{t-1}$ *are described by*

$$\Delta_1 = (F\beta_F)^2\sigma_z^{-2}; \qquad \Delta_2 = (N\beta_S)^2\sigma_z^{-2}.$$

□

Regarding the effect of speed competition, we conduct a numerical analysis to compare the equilibrium properties between the benchmark economy and the economy with high-frequency trading in Figure 5. Different from the benchmark model, fast traders can trade in both periods. Even though the *temporal fragmentation effect* of speed technology still exists in this section, the results show that the intensified speed competition monotonically increases the overall price informativeness, which is different from the benchmark model.

Through numerical analysis, Figure 5 shows that the fast traders shrink their trading aggressiveness within the high-frequency economy because they consider their profits in the second period. Fast traders also trade in the second period and hence trade less aggressively in the early period to preserve the information rent. This results in lower intermediate price informativeness compared to the benchmark case. However, within the high-frequency economy, the late-period trading aggressiveness increases. This is because, with all traders participating, the late-period market is liquid. Hence, compared to the benchmark economy, the overall price informativeness improves.

Finally, with high-frequency traders, the overall price informativeness monotonically increases. This is because speed competition does not reduce the number of slow traders resolving the long-run price discovery due to all traders trading in the late period. At the same time, as the speed competition becomes more intensified, more traders concentrate on the early period, which also leads to less pre-trading uncertainty for the late period. Hence, speed competition benefits price informativeness.

[Insert Figure 5 here]

6. Conclusions

Technology facilitates fast trading and influences information aggregation. Drawing on this insight, this paper studies a speed competition model with large strategic investors. Large traders internalize the price impact of their trades and strategically shade trading aggressiveness to resolve a fraction of the pre-trading uncertainty proportional to the number of traders in each trading round. Therefore, the concentration of investors' speed choice harms the overall information aggregation, generating hump-shaped price informativeness to speed competition.

With fast information diffusion, speed competition harms the overall price informativeness. When fast traders are high-frequency traders who trade in both periods, speed competition benefits the overall price informativeness.

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Table 1: Speed characteristics in sequential trading models

	Investor type	Early period trade	Late period trade	Competitive (C) /Strategic (S)	Endogenous type choice	Bundling information and speed
Hirshleifer et al. (1994)	Early informed	√	√	C		√
	Late informed	√	√			
Banerjee et al. (2018)	Long-lived traders	√	√	C		√
	Short-lived traders		√			
Froot et al. (1992)	Early informed	√		S		√
	Late informed		√			
Holden and Subrahmanyam (1996)	Long-horizon investors	√	√	C	√	√
	Short-horizon investors	√	Close out			
Dugast and Foucault (2018)	'Raw' signal investors	√		C	√	√
	'Processed' signal investors		√			
Kendall (2018)	'Rush' trader	√	√	C	√	√
	'Wait'' trader		√			
Huang and Yueshen (2021)	Fast trader	√		C	√	
	Slow trader		√			

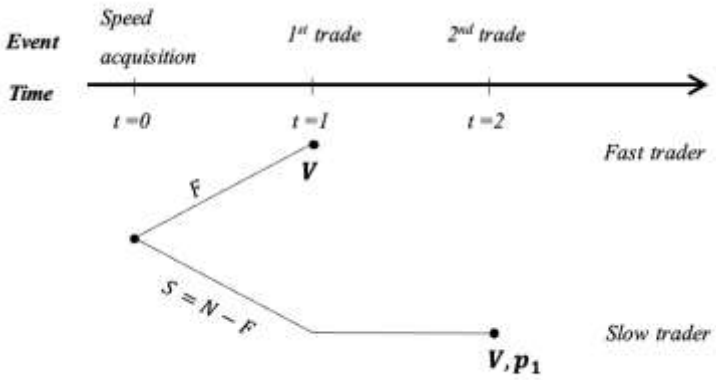


Figure 1 (a): Timeline for the benchmark model

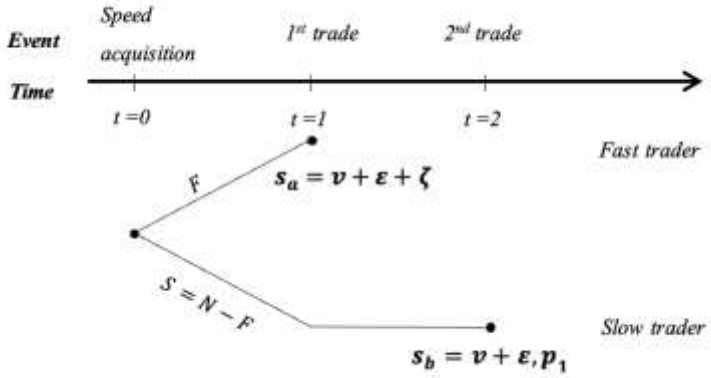


Figure 1 (b): Timeline for speed competition with information diffusion

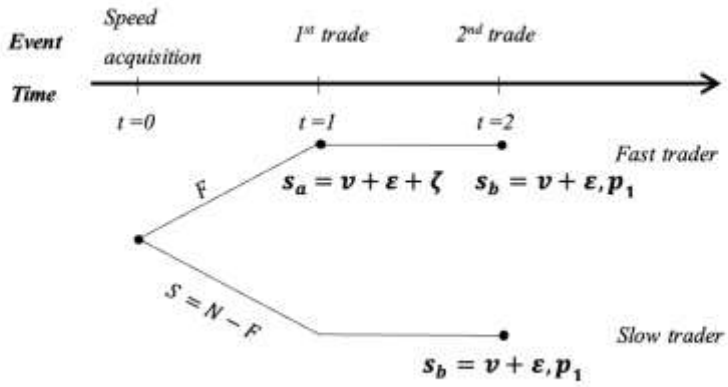


Figure 1 (c): Timeline for high-frequency trading with information diffusion

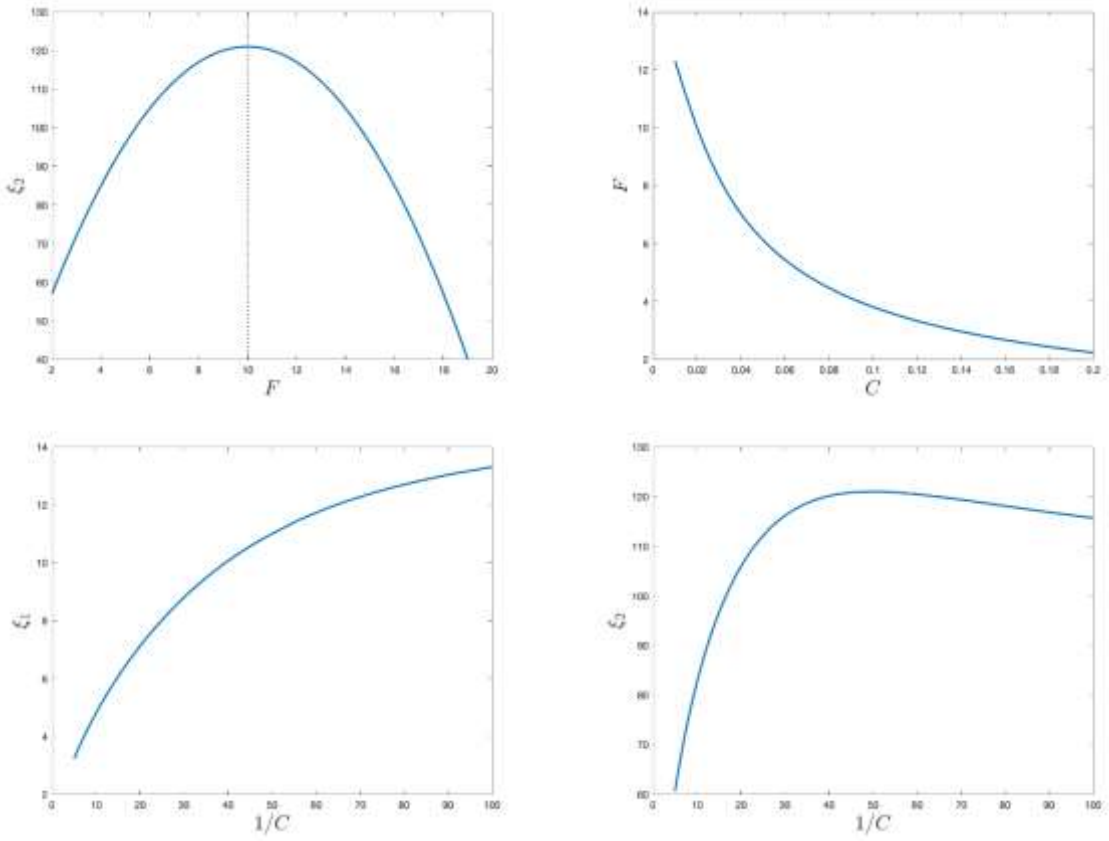


Figure 2: Equilibrium Price Informativeness for the Benchmark Model

In the benchmark economy, panel (a) describes how the overall price informativeness changes regarding more intensified speed competition as indicated by more fast traders. Panel (b) describes how the number of fast traders varies with speed acquisition cost. Panels (c) and (d) illustrate how intermediate and overall price informativeness change with more advanced speed technology. The parameters are: $\sigma = \sigma_z = 1$ and $N = 20$.

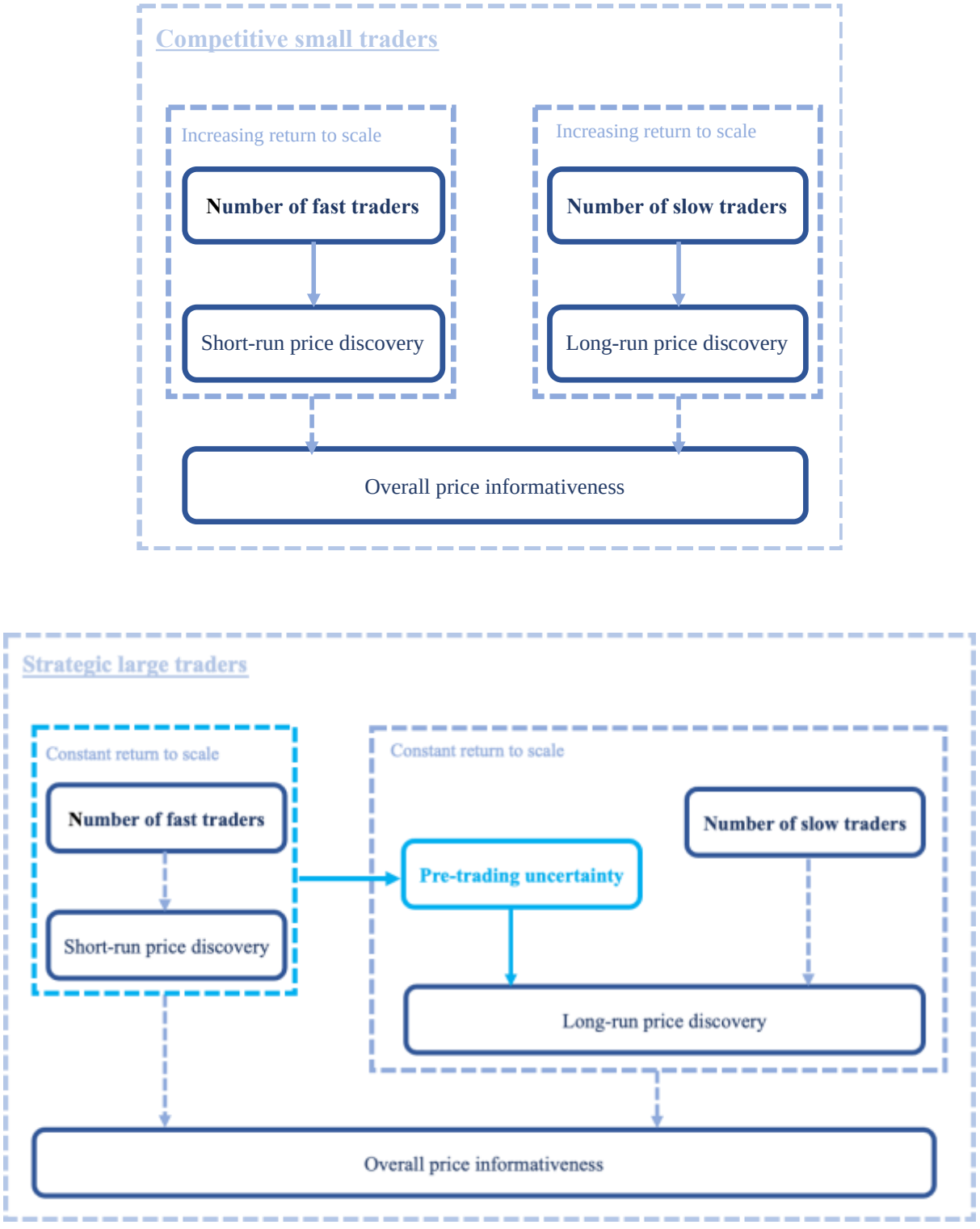


Figure 3: The temporal fragmentation effect of strategic trading compared to competitive trading

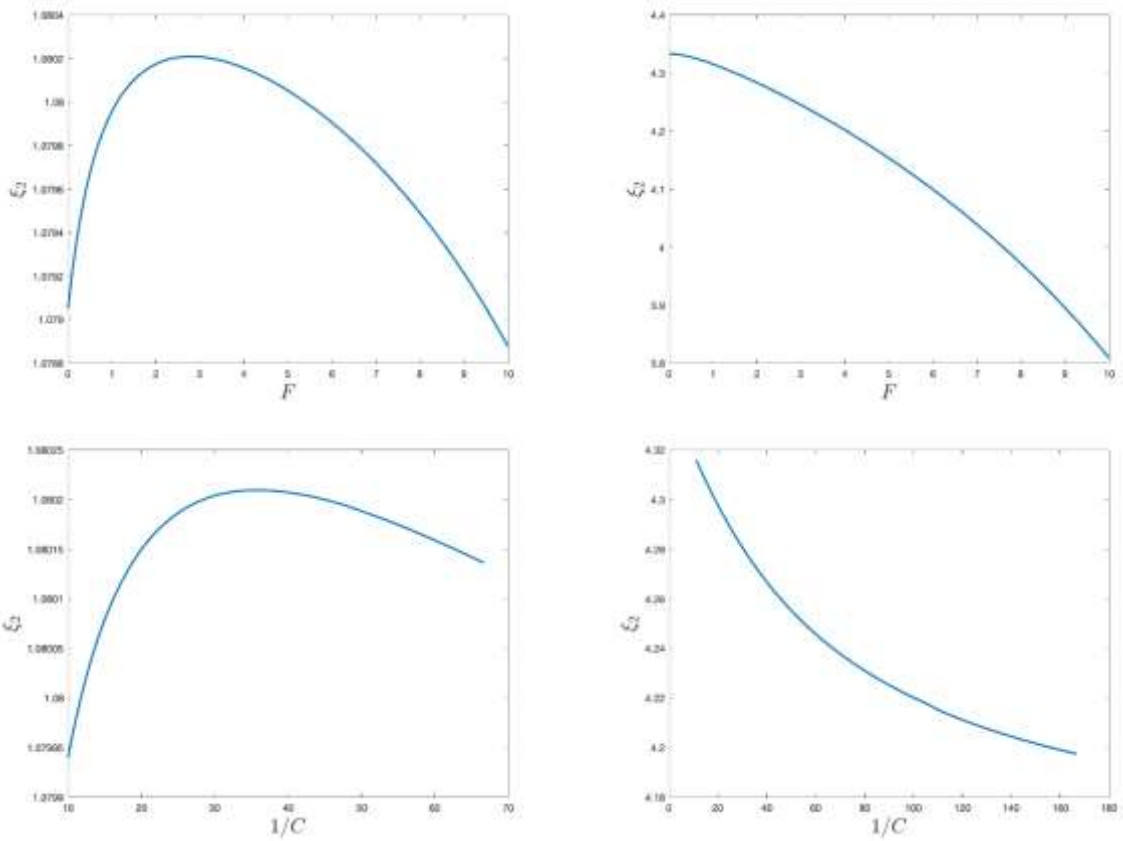


Figure 4: Equilibrium Price Informativeness with Information Diffusion

In the economy with information diffusion, panels (a) and (b) describe how the overall price informativeness changes regarding more intensified speed competition as indicated by more fast traders with different information diffusion $\eta = 1$ and $\eta = 20$. Panels (c) and (d) illustrate how the overall price informativeness change regarding more advanced speed technology with different information diffusion $\eta = 1$ and $\eta = 20$. The parameters are: $\sigma = \sigma_z = 1,\ \sigma_f = 5$ and $N = 20$.

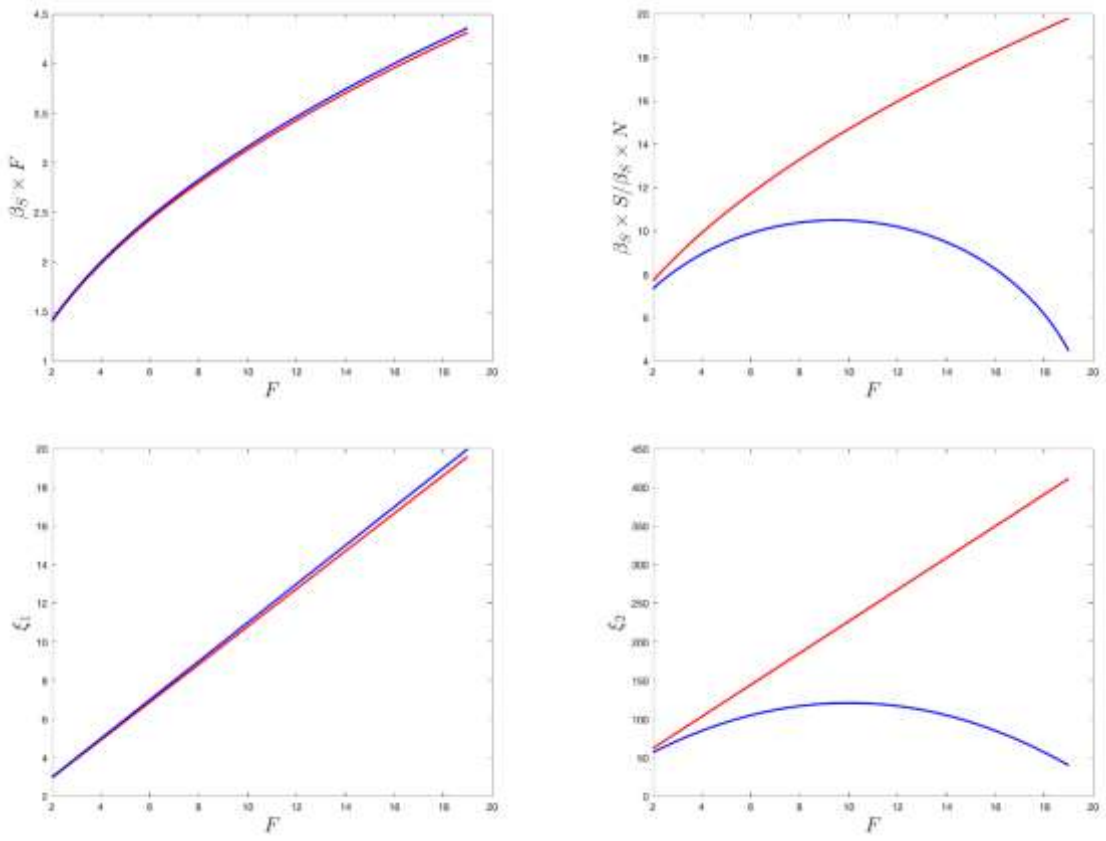


Figure 5: Comparison between the Benchmark Economy and the Economy with High-Frequency Trading

This figure compares the benchmark economy (blue lines) and the economy with high-frequency trading when fast traders trade in both periods (red lines). Panels (a) – (d) illustrate the aggregate trading aggressiveness in the early period and late period, intermediate and overall price informativeness, respectively. The parameters are: $\sigma = \sigma_z = 1$ and $N = 20$.

APPENDIX: PROOFS AND DISCUSSIONS

This appendix provides all the proofs and further discussions.

A1. Proofs of Lemma1 and Lemmas 3

We focus on the proofs of Lemma 3 since Lemmas 1 can be treated as a special situation of Lemma 3. Because fast traders only trade at the early period, we first calculate the equilibrium in the early period and then turn to the late period. With the speed hierarchies and the first-mover advantage, fast trader's trading behaviour is not influenced by slow traders. The equilibrium in the early period is thus similar to the standard one-period Kyle (1985) model. The standard procedure delivers price impact and fast traders' trading strategy as follows,

$$\lambda_1 = \frac{\sqrt{F}}{F+1}\frac{\sigma^2}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}\frac{1}{\sigma_z}; \qquad \beta_F = \frac{1}{\sqrt{F}}\frac{\sigma_z}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}. \tag{A-1}$$

With $F = 1$ and $\sigma_\varepsilon^2 = \sigma_\zeta^2$, (A-1) reduces to the standard one-period informed strategic trading scenario as in Kyle (1985).

We now turn to the equilibrium in the late period. With the information diffusion process, slow traders process better information. Based on the market maker's linear pricing functions, we first solve the speculator's optimal investment problem. With the speculator's strategy, we then determine the pricing functions of the market maker. On the one hand, since slow traders only trade in the late period and fast traders only trade in the early period, the slow traders' trading strategy $x_S(s_b, p_1) = \beta_s s_b - kp_1$ mirrores the standard one-period Kyle (1985),

$$\beta = \frac{1}{\lambda_2(S+1)}\frac{\sigma^2}{\sigma^2+\sigma_\varepsilon^2}; \qquad k = \frac{1}{\lambda_2(S+1)}.$$

On the other hand, the market maker takes the slow traders' trading strategy as given. The aggregate trading volume is given by

$$w_2 = Sx_S + z_2.$$

Substituting the expression of the demand functions into the aggregate trading volume, we have

$$w_2 = \frac{S}{\lambda_2(S+1)}\frac{\sigma^2}{\sigma^2+\sigma_\varepsilon^2}s_b - \frac{S}{\lambda_2(S+1)}p_1 + z_2.$$

In the late period, the market maker observes the aggregate trading volumes in both periods. Applying the projection theorem, we have

$$\lambda_2 = \frac{Cov(V, w_2)Var(w_1) - Cov(w_2, w_1)Cov(V, w_1)}{Var(w_2)Var(w_1) - Cov(w_2, w_1)^2}.$$

From this, we obtain the equilibrium conditions for the late period. In equilibrium, the price impact and the slow traders' trading strategy follow

$$\lambda_2 = \sqrt{\frac{(F+1)\sigma^2\sigma_\zeta^2 + (\sigma^2+\sigma_\varepsilon^2)\sigma^2}{(\sigma^2+\sigma_\varepsilon^2)(\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2)}}\sqrt{\frac{S}{F+1}}\frac{1}{S+1}\frac{\sigma}{\sigma_z}; \tag{A-2}$$

$$\beta_S = \frac{\sqrt{(\sigma^2+\sigma_\varepsilon^2)(\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2)}}{\sqrt{(F+1)\sigma^2\sigma_\zeta^2 + (\sigma^2+\sigma_\varepsilon^2)\sigma^2}}\sqrt{\frac{F+1}{S}}\frac{\sigma^2}{\sigma^2+\sigma_\varepsilon^2}\frac{\sigma_z}{\sigma}; \qquad k = \frac{\sigma^2+\sigma_\varepsilon^2}{\sigma^2}\beta_S. \tag{A-3}$$

Equations (A-1) to (A-3) describe the financial market equilibrium in Lemma 3. The expected trading profit and price informativeness are then calculated. Note that Lemma 1 is a special case of Lemma 3 with parameter value $\sigma_\varepsilon^2 = \sigma_\zeta^2 = 0$.

The endogenous number of fast traders is closely related to the difference between fast and slow traders' expected profits. To explore the endogenous equilibrium properties, we denote the expected profits difference function by $\mathcal{G}(F)$,

$$\mathcal{G}(F) = \left[\frac{1}{(F+1)\sqrt{F}}\frac{\sigma}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}\right]\sigma\sigma_z - \left[\frac{1}{(S+1)\sqrt{S}}\sqrt{\frac{\sigma_\zeta^2}{\sigma^2+\sigma_\varepsilon^2}+\frac{1}{F+1}}\frac{\sigma}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}\right]\sigma\sigma_z.$$

Calculating the first order derivative for function $\mathcal{G}(F)$, we have:

$$\frac{d\mathcal{G}(F)}{dF} = \frac{\sigma^2\sigma_z}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}\left[-\frac{2F+1}{2(F+1)^2F\sqrt{F}} + \frac{2S+1}{2(S+1)^2S\sqrt{S}}\sqrt{\frac{\sigma_\zeta^2}{\sigma^2+\sigma_\varepsilon^2}+\frac{1}{F+1}} - \frac{1}{\sqrt{S}(S+1)}\frac{1}{2}\frac{\frac{1}{(F+1)^2}}{\sqrt{\frac{\sigma_\zeta^2}{\sigma^2+\sigma_\varepsilon^2}+\frac{1}{F+1}}}\right] < 0. \tag{A-4}$$

This means that the difference between the fast and slow traders' expected profits decreases with more fast traders. At the same time, it is easy to see that

$$\mathcal{G}(N-1) = \frac{\sigma^2\sigma_z}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}\left[\frac{1}{N\sqrt{N-1}} - \frac{1}{2}\sqrt{\frac{\sigma_\zeta^2}{\sigma^2+\sigma_\varepsilon^2}+\frac{1}{N}}\right] < 0.$$

This means that not every trader rushes to be fast due to strategic price impact even without speed cost. Hence, within the strategic setup, the speed heterogeneity endogenously arises. Furthermore, to ensure there is at least one fast trader, we impose the following condition,

$$\mathcal{G}(1) = \frac{\sigma^2\sigma_z}{\sqrt{\sigma^2+\sigma_\varepsilon^2+\sigma_\zeta^2}}\left[\frac{1}{2} - \frac{1}{\sqrt{(N-1)N}}\sqrt{\frac{\sigma_\zeta^2}{\sigma^2+\sigma_\varepsilon^2}+\frac{1}{2}}\right] > 0,$$

and the speed cost must satisfy

$$C \le \hat{C} = \frac{\sigma^2 \sigma_z}{\sqrt{\sigma^2 + \sigma_\varepsilon^2 + \sigma_\zeta^2}} \left[\frac{1}{2} - \frac{1}{\sqrt{(N-1)N}} \sqrt{\frac{\sigma_\zeta^2}{\sigma^2 + \sigma_\varepsilon^2}} + \frac{1}{2} \right].$$

Otherwise, all traders choose to stay slow.

A2. Proof of Lemma 2

Based on the definition of Nash equilibrium, no one deviates from their equilibrium strategy. Hence, for given speed cost C, we have the market equilibrium fraction of the fast traders F^* satisfying

$$\pi_F(F^*, N - F^*) - C \ge \pi_S(F^* - 1, N - F^* + 1);$$
$$\pi_S(F^*, N - F^*) \ge \pi_F(F^* + 1, N - F^* - 1) - C,$$

for $F^* \ge 1$ or $\Gamma = \pi_F - \pi_S < C$ for $F^* = 0$. This means that the fast trade does not switch back to the slow, and vice versa. Based on the break-even condition between the fast and slow traders' expected profit,

$$\mathcal{G}(X) = \frac{1}{(X+1)\sqrt{X}} \sigma\sigma_z - \frac{1}{(N - X + 1)\sqrt{N - X}\sqrt{X + 1}} \sigma\sigma_z - C = 0,$$

we can have that X renders the fast to be indifferent from the slow. However, this number is not necessarily an integral. The strict definition of the equilibrium instead requires the number of fast traders to be integral. We first investigate the properties of fast and slow traders' expected profit separately, which is crucial for determining the equilibrium. We can show that the fast trader's expected profit decreases with the number of fast traders, while the change of slow trader's expected profit to number of fast traders can be reflected by

$$\frac{d}{dF}\left[\frac{1}{(S+1)\sqrt{S}\sqrt{F+1}} \sigma\sigma_z \right] \propto -4S^2 + (3N+1)S + N + 1.$$

This means that when

$$X < \bar{X} = \frac{5N - 1 - \sqrt{(3N+1)^2 + 16(N+1)}}{8},$$

The slow trader's expected profit decreases with number of fast traders. Also, when $X \ge \bar{X}$, the slow trader's expected profit increases with the number of the fast traders. The equilibrium then depends on which regime X is located.

<u>*Scenario 1*: $X \ge \bar{X}$</u>. We first check whether $\lceil X \rceil > X$ is an equilibrium solution. Under this scenario, the slow trader has no incentive to deviate from the equilibrium,

$$\pi_S(\lceil X \rceil, N - \lceil X \rceil) > \pi_F(\lceil X \rceil, N - \lceil X \rceil) - C > \pi_F(\lceil X \rceil - 1, N + 1 - \lceil X \rceil) - C.$$

The left side inequality is due to the ceiling calculation while the right side inequality is due to the decreasing expected profit of the fast trader when $X \geq \bar{X}$. However, the fast trader can deviate from the equilibrium due to

$$\pi_S(X, N-X) = \pi_F(X, N-X) - C > \pi_F(\lceil X \rceil, N - \lceil X \rceil) - C,$$
$$\pi_S(X, N-X) = \pi_F(X, N-X) - C > \pi_S(\lfloor X \rfloor, N - \lfloor X \rfloor).$$

We cannot deterministically find the lager value between $\pi_F(\lceil X \rceil, N - \lceil X \rceil) - C$ and $\pi_S(\lfloor X \rfloor, N - \lfloor X \rfloor)$. Therefore, the equilibrium solution depends on these two values. Specifically, when $\pi_F(\lceil X \rceil, N - \lceil X \rceil) - C > \pi_S(\lfloor X \rfloor, N - \lfloor X \rfloor)$, the fast trader also does not deviate from the equilibrium. By contrast, when $\pi_F(\lceil X \rceil, N - \lceil X \rceil) - C \leq \pi_S(\lfloor X \rfloor, N - \lfloor X \rfloor)$, the fast deviates from the equilibrium to $\lfloor X \rfloor$. Now the equilibrium number of fast traders becomes $\lfloor X \rfloor$. Furthermore, under this situation, the slow trader does not switch to the fast due to the same condition and fast does not switch to slow because that the difference between the fast and slow traders' expected profit is decreasing with respect to number of the fast traders.

Scenario 2: $X < \bar{X}$. We first show that $\lceil X \rceil$ is not an equilibrium solution. We can show that slow trader is definitely deviate from the equilibrium when

$$\pi_S(\lfloor X \rfloor, N - \lfloor X \rfloor) > \pi_S(X, N-X) = \pi_F(X, N-X) - C > \pi_F(\lceil X \rceil, N - \lceil X \rceil) - C.$$

The first inequality is due to that the value of X is located in the range in which the expected profit of the slow trader decreases in the number of the fast traders. The second inequality is due to the decreasing profit of the fast traders. Hence, the equilibrium can only be smaller than X. Under this condition, the slow does not change to the fast. Therefore, we can focus on the fast decision. When $\pi_F(\lfloor X \rfloor, N - \lfloor X \rfloor) - C > \pi_S(\lfloor X \rfloor - 1, N + 1 - \lfloor X \rfloor)$, the fast trader also does not deviate from $\lfloor X \rfloor$. By contrast, when $\pi_F(\lfloor X \rfloor, N - \lfloor X \rfloor) - C \leq \pi_S(\lfloor X \rfloor - 1, N + 1 - \lfloor X \rfloor)$, the fast deviates from $\lfloor X \rfloor$. This completes the proof of Lemma 2.

A3. Proof of Proposition 2

From Proposition 1, the overall price informativeness ξ_2 improves initially but reduces eventually, exhibiting a hump-shape relationship with a maximum at $\bar{F} = N/2$, *i.e.*, $\partial \xi_2 / \partial F > 0$ (< 0) *for* $F < N/2 (> N/2)$. On the other hand, substituting $\bar{F} = N/2$ into the difference between the fast and slow trader's expected profit, we have

$$\frac{1}{\left(\frac{N}{2}+1\right)\sqrt{\frac{N}{2}}} \sigma \sigma_z \left(1 - \frac{1}{\sqrt{\frac{N}{2}+1}}\right) > 0.$$

This means that, without speed cost, the equilibrium number of the fast traders is higher than $\bar{F}$. Therefore, we can define a speed cost that renders the equilibrium number of fast traders equal to $\bar{F}$. Furthermore, considering the decreasing difference between the fast and slow traders' expected profit (as in (A-4)), we have Proposition 2.

A4. Proof of Proposition 3 and Proposition 4

Based on the expression of overall price informativeness, the number of fast traders influences the price informativeness through

$$\frac{d\xi_2}{dF} \propto -\frac{d}{dF}\left[\frac{\sigma^2}{S+1}\left(\sigma_\zeta^2+\frac{\sigma^2+\sigma_\varepsilon^2}{F+1}\right)\right] = -\frac{\sigma^2}{(S+1)^2(F+1)^2}\left[\sigma_\zeta^2F^2+2\left(\sigma_\zeta^2+\sigma^2+\sigma_\varepsilon^2\right)F+\sigma_\zeta^2-(\sigma^2+\sigma_\varepsilon^2)N\right].$$

Hence, the impact of speed competition on price informativeness depends on

$$-\sigma_\zeta^2F^2-2\left(\sigma_\zeta^2+\sigma^2+\sigma_\varepsilon^2\right)F-\sigma_\zeta^2+(\sigma^2+\sigma_\varepsilon^2)N.$$

The above expression is a quadratic function of F. When $\sigma_\zeta^2 > (\sigma^2+\sigma_\varepsilon^2)N$, the above quadratic function is always positive in the range $F\epsilon[1, N-1]$. By contrast, when $\sigma_\zeta^2 < (\sigma^2+\sigma_\varepsilon^2)N$, the above quadratic function can have negative value. Hence, under this scenario, the overall price informativeness ξ_2 improves initially but reduces eventually, i.e., $\partial\xi_2/\partial F > 0$ (< 0) *for* $F < \hat{F}(> \hat{F})$, where

$$\hat{F} = \frac{\sqrt{1+(2+N)\eta}-(1+\eta)}{\eta}.$$

Furthermore, the marginal number of fast traders $\ddot{F}$ rendering the expected profit of the fast and slow traders to be equal satisfies

$$(\eta+1)\ddot{F}^3+(2\eta-3N-1)\ddot{F}^2+\left(\eta+1+(3N+1)(N+1)\right)\ddot{F}-N(N+1)^2=0.$$

Substituting $\hat{F}$ into the above function, we have

$$(\eta+1)\hat{F}^3+(2\eta-3N-1)\hat{F}^2+\left(\eta+1+(3N+1)(N+1)\right)\hat{F}-N(N+1)^2>0.$$

This means that $\hat{F} < \ddot{F}$. Considering that the difference between the fast and slow traders' expected profit is monotonically decreasing, we have Proposition 3 and Proposition 4.